

Chapter 1 – Visual C# Concept Notes (1–14)

1. Object

An object is a program component that contains data and performs operations.

Property = data stored in an object.

Method = an operation an object can perform.

2. Controls

Controls are objects that are visible in a GUI. Examples: Label, Button, TextBox.

3. Class

A class is code that describes a particular type of object.

4. .NET

.NET is a collection of classes and other code that can be used to create programs for window operating systems.

5. Visual Studio

Visual Studio is a professional IDE (Integrated Development Environment).

Important windows include Designer, Solution Explorer, Properties.

6. Project vs. Solution

Project = one application.

Solution = a container that can hold one or more projects.

Project files = the code and resources used by the application.

7. Form

A Form is the main window/interface of a Windows Forms application.

8. Properties

Properties are settings that control how an object looks and behaves.

9. Control Naming

A control name must start with a letter or underscore. Other characters can be letters, numbers, or underscores. Spaces are not allowed. The chapter presents camelCase as a common naming convention.

10. Namespace fi Class fi Method

Namespace = container that holds classes.

Class = container that holds methods.

Method = a group of statements that performs an operation.

11. Event-Driven Programming

A GUI application responds to events such as a button click, key press, or mouse action.

12. Event Handler

An event handler is a method that executes when a specific event takes place.

13. MessageBox

Displays a message to the user.

14. Label

A Label displays text on a form and can also display program output. Important properties include Text, Name, Font, BorderStyle, AutoSize, and TextAlign.